BRAIN ASTROCYTOMAS

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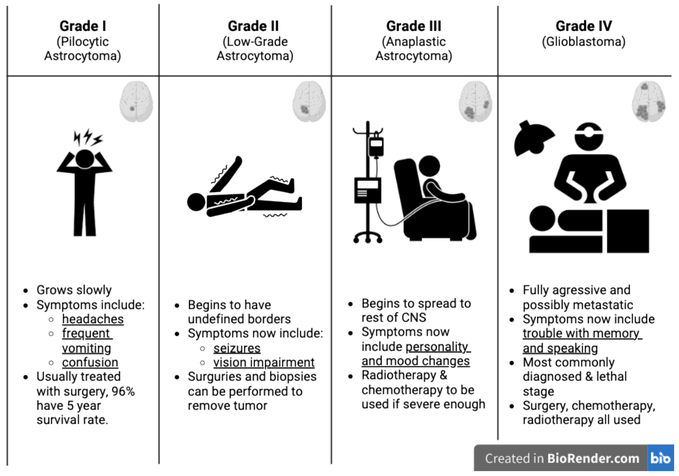
ABSTRACT

Astrocytoma is the most common and deadly type of brain cancer, with 15,000 new cases being diagnosed every day. The way astrocytoma develops is complex, involving the complex interactions between the intercellular and intracellular pathways. Due to astrocytoma's often late-stage diagnosis in the form of glioblastoma, curing it and finding possible treatments for it is very difficult. However, by fully analyzing the mechanisims of astrocytoma, many different potential treatment options can be indetified for future research.

INTRODUCTION

Astrocytoma originates from astrocytes, which are cells that act as guides for other cells in the brain by promoting chemical signaling. These tumors usually originate in the brain and spinal cord, which can easily spread to the rest of the central nervous system if it metastisizes. Six different chemical interactions and pathways have been identified for the development of astrocytoma: a mutated *BRAF* gene, the overexpression of epidermal growth factor receptor (EGFR), the loss of function og Caspace 9 and 3, telomeric extension by cancer-specific APB, angiogensis caused by VEGF stimulation, and the EMT process in some tumor cells to cause metastasis.

Astrocytomas have been shown to be one of the most diffucult cancers to treat and research due to its most diagnosed form, gliblasotma, being very late-stage. However, astrocytoma can be diagnosed and treated very early with proper screening tests and equipment, and better treatment methods are being developed today to target the multitude of astrocytoma genesis pathways.



RESULTS

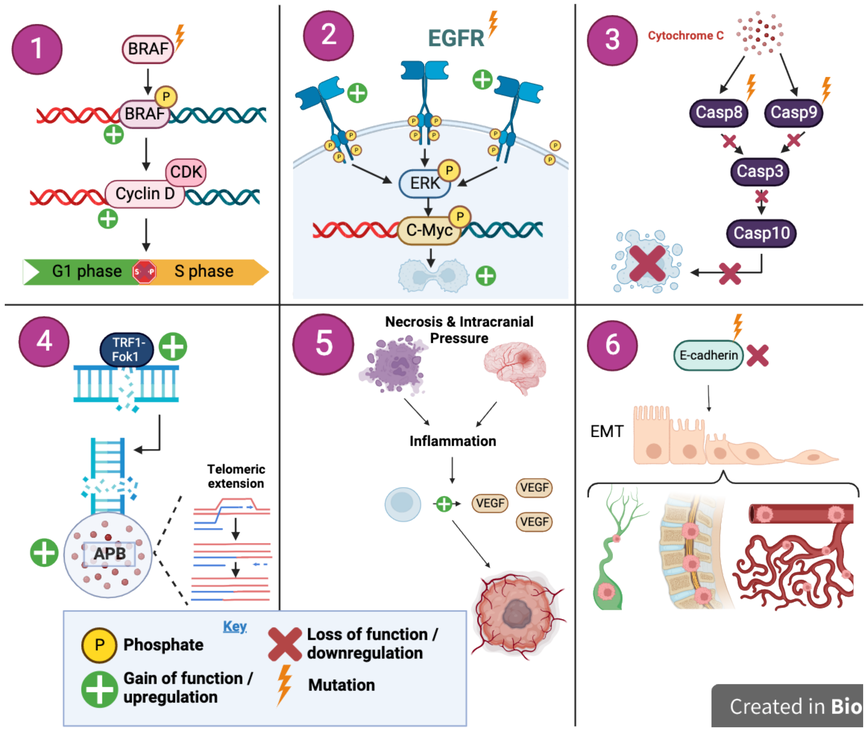


Figure 1: 1. Evading Growth Supressors, 2. Sustaining Proliferative Growth, 3. Resisting Cell Death, 4. Enabling Replicative Immortality, 5. Inducing Angiogenesis, 6. Activating Invasion and Metastasis

RISK FACTORS

There are many major risk factors that can be associated with astrocytomas in the brain. If the mutations that cause the astrocytomas are germline mutations, then there are multiple hereditary syndromes associated with astrocytomas. This includes tuberous sclerosis complex (TSC), Li-Fraumeni syndrome, Turcot syndrome, nevoid basal cell carcinoma syndrome (NBCCS), and constitutional mismatch repair deficiency syndrome. TSC has mutations in two genes, TSC1 and TSC2, which are involved in stopping cell proliferation and stopping cancer from occurring. Li-Fraumeni syndrome is caused by TP53 mutations, which are involved in stopping cell division. Turcot syndrome is a form of Lynch syndrome, and it is caused by mutations in APC gene. Mutations in APC can increase the risk for colorectal polyps and polyps in other body parts like the brain.

CONCLUSION AND FUTURE DIRECTIONS

The physiological development of astrocytomas has been demostrated to be very complex in its interconnected hallmarks, signaling pathways and in its limited treatment options due its late-stage diagnosis. However, a greater picture of its prognosis has allowed us to understand current reseach being develop to treat astrocytoma in all its stages and postulate future targets for treatment.

In the future, research for astrocytomas is bound to happen to find new treatment options. One major option that research is being done is genetic therapies using CRISPR-Cas9 systems to target glioblastomas using gRNAs and nanocapsules. Also, research is also going on related brain tumors like pineoblastomas to design genetic therapies for them as a form of personalized medicine as well.

OUTCOMES

There are three treatment methods that can be used to cure astrocytomas. The outcome behind surgery is that the tumor will be removed as closely as possible without any damage to healthy cells. In radiation, the outcome is the removal of the tumor at the specific site through damage of DNA, preventing the tumor cells from replicating and forcing them to die. In chemotherapy, the outcome is the death of the tumor through the usage of drugs that target various hallmarks of cancer. These treatments are the most cutting edge treatments that have been used and perfected as of now.

DIAGNOSIS, TREATMENT, AND PREVENTION

Astrocytomas are mainly diagnosed throguh MRI's and CT scans. Common treatments include surgery, radiation therapy, and chemotherapy. The most common chemotherapies are Avastin, which targets angiogenesis throguh VEGF, and Temozolomide, which targets cell proliferation by preventing cells from making DNA. There are no known major environmental or lifestyle-related factors that can decrease the chance of getting astrocytoma other than avoiding exessive radiation exposure.

REFERENCES AND ACKNOWLEDGEMENTS

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